CYBERSECURITY INCIDENT REPORT

Simulated Windows / SIEM Log Investigation

Incident ID: SIM-2026-0921-01 **Date:** 21 September 2026

Classification: Simulated / Academic Exercise

Executive Summary

The simulated SIEM data indicates a possible compromised administrative credential followed by privilege use, encoded PowerShell execution, reconnaissance, suspicious outbound communication, DLL execution, and creation of a privileged persistence account.

Detected Anomalies

ID	Anomaly	Evidence	Severity
A01	Repeated failed logons followed by success	3x Event 4625 then 4624 for Administrator from 10.0.5.23	High
A02	Privileged logon	Event 4672 for Administrator from workstation IP 10.0.5.23	High
A03	Encoded PowerShell	Event 4688 shows powershell.exe -enc	Critical
A04	Reconnaissance commands	whoami and ipconfig executed remotely	Medium
A05	Suspicious outbound connection	10.0.5.23 → 185.199.110.153:443	High
A06	Suspicious DLL execution	rundll32.exe suspicious.dll,Start	Critical
A07	New privileged account	svc_backup created and added to Administrators	Critical

Indicators of Compromise

• IP address: 10.0.5.23 — suspicious source of privileged logons and process creation.

• External IP: 185.199.110.153:443 — suspicious outbound connection observed from FILE01.

• Account: svc_backup — newly created account subsequently added to local Administrators.

• Command pattern: powershell.exe -enc — encoded PowerShell execution.

• Executable pattern: rundll32.exe suspicious.dll,Start — suspicious DLL execution.

• Potential artifact: suspicious.dll — filename supplied by the simulated log; file hash is not available.

Response and Containment

1. Isolate FILE01 from the network while keeping the system powered on if forensic procedures require volatile evidence.

2. Disable or reset the affected Administrator credentials after preserving relevant authentication evidence.

3. Disable svc_backup and remove it from privileged groups after documenting the account state.

4. Block the suspicious external destination at firewall/proxy controls, subject to incident-response evidence requirements.

5. Preserve Windows Security logs, PowerShell logs, Sysmon/EDR telemetry, firewall/proxy logs, and relevant endpoint artifacts.

6. Collect and hash suspicious.dll and other newly created or modified files; obtain a memory image if available.

7. Review other systems for logons from 10.0.5.23, use of svc_backup, the suspicious DLL, and the external IP.

8. Reset potentially exposed credentials and enforce MFA/privileged-access controls where available.

9. After eradication, patch affected systems, remove persistence, restore from a trusted state if required, and monitor for recurrence.

Conclusion

Treat the simulated case as a high-priority incident until disproven. Preserve evidence, isolate the suspected endpoint, contain credentials and persistence, block suspicious communication, and hunt across the environment for the same indicators.